Entropy as Loss of Accessible Coherence in Darke Theory

Why Entropy Is Not Primitive Disorder

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Abstract

Darke Theory rejects the treatment of entropy as a primitive measure of disorder. Disorder language is descriptive at best, coarse-graining dependent, and structurally underdetermined. This paper develops a mathematically stronger reconstruction in which entropy is defined as loss of accessible coherence under recursive differentiation, closure constraint, support limitation, and admissibility loss. A total coherence functional is introduced and decomposed into internal, relational, closure-bearing, and residual components. Accessibility is then formalised as a structural filter governing how much of that coherence remains navigable, supportable, recoverable, and operationally activatable. Accessible coherence and closure-accessible coherence are defined from these objects, and Darke entropy is introduced as a monotone decreasing transform of closure-accessible coherence. This yields a structurally precise criterion: entropy increases if and only if closure-accessible coherence decreases. The framework then proves that entropy may rise even when total coherence is preserved or increased, thereby separating entropy from naive disorder. Irreversibility is formalised through recovery burden and recovery reserve, allowing entropy increase to be identified with coherence loss that cannot be restored admissibly. Classical thermodynamic entropy is then reinterpreted as a coarse-grained macrostatistical projection of this deeper coherence-accessibility law. Finally, entropy increase is related to structural memory growth without collapsing the two concepts into identity. The result is a Darke-Theoretic entropy law in which entropy is not primitive disorder but the inverse image of persistent accessible coherence under structural constraint.

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1 Introduction

Entropy is one of the most frequently invoked and least carefully grounded concepts in physics. Standard exposition often treats entropy as disorder, spread, randomness, or multiplicity. Such language can be useful pedagogically, but it is not foundationally adequate. Disorder is not a primitive structural object. It depends on description level, coarse-graining choice, observational resolution, and implicit assumptions about what counts as relevant organisation. A concept so dependent on descriptive framing cannot be the deepest ontological meaning of entropy.

Darke Theory requires a more disciplined reconstruction.

The framework does not begin from thermodynamic folklore, nor does it treat entropy as an independent primitive. It begins from recursive differentiation, closure, admissibility, and persistent structure. Once that architecture is taken seriously, the correct question is not *how disordered is a state*, but rather:

> *How much structurally usable coherence remains accessible under the closure, burden, support, and recovery constraints of the system?*

That question leads to a different entropy law.

The central claim of this paper is:

> **Entropy in Darke Theory is the loss of accessible coherence.**

More strongly, the present paper argues that entropy should be defined as a monotone decreasing functional of closure-accessible coherence. This immediately yields several consequences:

1. entropy increase is not identical to visual or descriptive disorder,
2. entropy may increase even when total coherence remains large,
3. irreversibility is linked to inadmissible recovery of lost accessible coherence,
4. thermodynamic entropy becomes a regime-level projection rather than the deepest ontology.

The purpose of this paper is therefore not merely to relabel standard entropy. It is to reconstruct the concept from the load-bearing primitives of Darke Theory.

2 Foundational discipline

The present construction stands downstream of the following structural commitments.

Principle 1 (No primitive entropy)**.** Entropy may not be treated as a primitive ontological measure if it is to be explained in terms of the conditions under which structure, persistence, and recoverability become meaningful.

Principle 2 (Coherence priority)**.** In Darke Theory, coherence is more primitive than entropy. Entropy must therefore be reconstructed from the fate of coherence under recursive differentiation and admissibility constraint.

Principle 3 (Accessibility priority)**.** Not all coherence present in a state is structurally usable. The entropy-relevant quantity is not mere coherence possession, but coherence that remains accessible under closure, support, burden, and recovery conditions.

Principle 4 (Admissible irreversibility)**.** Entropy increase becomes structurally significant when lost accessible coherence cannot be restored within the admissible sector of the governing system.

These principles jointly force a departure from disorder language. The correct foundational object is not disorder but accessible coherence.

3 Notation summary

For convenience, the main symbols are collected here.

Symbol	**Meaning**
$\mathcal{X}$	State space of refinement configurations.
$\preceq$	Refinement or structural-precedence partial order on $\mathcal{X}$.
D	Recursive substantiation operator on $\mathcal{X}$.
$\mathsf{Cl}(X)$	Graded closure functional of state X.
$\mathsf{Cl}_{\mathrm{crit}}$	Critical closure threshold.
$\mathcal{C}_{\mathrm{crit}}$	Closed sector above the critical closure threshold.
$\mathsf{S}(X \to Y; \ell)$	Available support for transition $X \to Y$ across horizon ℓ.
$\mathsf{B}(X \to Y; \ell)$	Burden required for transition $X \to Y$ across horizon ℓ.
$\Pi(X \to Y)$	Reserve margin $\mathsf{S} - \mathsf{B}$ for transition $X \to Y$.
$\mathsf{B}_{\mathrm{rec}}(Y \to X; \ell)$	Burden required to recover coherence lost in passing from X to Y.
$\Pi_{\mathrm{rec}}(Y \to X)$	Recovery reserve for reverse restoration.
$X \rightsquigarrow_A Y$	Admissible transition from X to Y.

Symbol	Meaning
$\mathsf{K}(X)$	Total coherence present in state X.
$\mathsf{K}_{\mathrm{acc}}(X)$	Accessible coherence in state X.
$\mathsf{K}_{\mathrm{cl,acc}}(X)$	Closure-accessible coherence in state X.
$\mathsf{S}_D(X)$	Darke entropy functional of state X.
$\mathsf{M}(X)$	Structural memory functional of state X.
$\mathsf{Res}(X)$	Persistent structural residue in state X.
$\mathsf{Br}(X)$	Irreversible branch-loss or excluded possibility measure in state X.
$\Omega(X, Y; \ell)$	Directional asymmetry functional.
Φ	Strictly decreasing entropy transform.
Γ	Monotone closure-gating function on closure score.
$\mathsf{A}_c(X)$	Coherence accessibility operator.
$\mathcal{P}(X_0, X)$	Family of admissible paths from reference state X_0 to X.

4 Recursive refinement substrate

The entropy reconstruction inherits the same structural substrate used elsewhere in Darke Theory.

Definition 1 (Recursive refinement system)**.** A recursive refinement system is a triple

$$(\mathcal{X}, \preceq, \mathsf{D}),$$

where:

i. $\mathcal{X}$ is a class of refinement states,

ii. $\preceq$ is a partial order representing refinement precedence,

iii. $\mathsf{D} : \mathcal{X} \to \mathcal{X}$ is a recursive substantiation operator.

We assume that $(\mathcal{X}, \preceq)$ is directed-complete on the sectors relevant to the present construction and that D is monotone:

$$X \preceq Y \implies \mathsf{D}(X) \preceq \mathsf{D}(Y).$$

When required, we also assume Scott continuity:

$$\mathsf{D}\left(\sup_i X_i\right) = \sup_i \mathsf{D}(X_i)$$

for directed families $\{X_i\}$.

Definition 2 (Recursive chain)**.** For $X \in \mathcal{X}$, a recursive chain generated from X is any sequence

$$X,\ \mathsf{D}(X),\ \mathsf{D}^2(X),\ \ldots$$

or more generally any chain compatible with the refinement order.

This substrate gives structurally grounded succession, but not yet entropy. Entropy requires the additional concepts of closure, supportability, and coherence accessibility.

5 Closure, support, and admissibility

Definition 3 (Closure subfunctionals)**.** Let

$$I_{\text{self}},\ I_{\text{rec}},\ I_{\text{supp}},\ I_{\text{pert}} : \mathcal{X} \to [0,1]$$

denote the subfunctionals for internal self-integration, recursive invariance, support stability, and perturbation resistance respectively.

Definition 4 (Closure functional)**.** Define

$$\mathsf{Cl} : \mathcal{X} \to [0,\infty)$$

by

$$\mathsf{Cl}(X) = \alpha_c I_{\text{self}}(X) + \beta_c I_{\text{rec}}(X) + \gamma_c I_{\text{supp}}(X) + \delta_c I_{\text{pert}}(X), \qquad \alpha_c, \beta_c, \gamma_c, \delta_c > 0.$$

Lemma 1. *For every $X \in \mathcal{X}$,*

$$0 \le \mathsf{Cl}(X) \le \alpha_c + \beta_c + \gamma_c + \delta_c.$$

Proof. Each subfunctional lies in $[0,1]$, and each coefficient is strictly positive.

Definition 5 (Closed sector)**.** For a critical threshold $\mathsf{Cl}_{\text{crit}} > 0$, define

$$\mathcal{C}_{\text{crit}} := \{X \in \mathcal{X} : \mathsf{Cl}(X) \ge \mathsf{Cl}_{\text{crit}}\}.$$

Definition 6 (Support and burden)**.** For a candidate transition $X \to Y$ across persistence horizon ℓ, define support

$$\mathsf{S}(X \to Y; \ell) \in [0,\infty)$$

and burden

$$\mathsf{B}(X \to Y; \ell) \in [0,\infty).$$

Definition 7 (Reserve)**.** Define the reserve margin

$$\Pi(X \to Y) := \mathsf{S}(X \to Y; \ell) - \mathsf{B}(X \to Y; \ell).$$

Definition 8 (Admissibility)**.** Let $\pi_{\min} \ge 0$ and $\mathsf{Cl}_{\min} \ge 0$. Define

$$\mathcal{A}(X \to Y) := \mathbf{1}\Big[\Pi(X \to Y) \ge \pi_{\min} \ \wedge\ \mathsf{Cl}(Y) \ge \mathsf{Cl}_{\min}\Big].$$

Write

$$X \rightsquigarrow_A Y$$

if and only if $\mathcal{A}(X \to Y) = 1$.

Remark 1. Reserve subsumes raw burden feasibility. If $\Pi(X \to Y) \ge \pi_{\min}$ with $\pi_{\min} \ge 0$, then the burden cannot exceed available support.

This admissibility layer matters because entropy increase is not merely any loss of coherence. It is loss of coherence that is no longer accessible or recoverable on the admissible sector.

6 Total coherence

We now introduce the first genuinely entropy-specific object.

Definition 9 (Coherence components)**.** Let

$$\mathsf{K}_{\mathrm{int}},\ \mathsf{K}_{\mathrm{rel}},\ \mathsf{K}_{\mathrm{cl}},\ \mathsf{K}_{\mathrm{res}} : \mathcal{X} \to [0,1]$$

denote the internal, relational, closure-bearing, and residual coherence components respectively.

Definition 10 (Total coherence)**.** Define the total coherence functional

$$\mathsf{K} : \mathcal{X} \to [0,\infty)$$

by

$$\mathsf{K}(X) = \alpha_1 \mathsf{K}_{\mathrm{int}}(X) + \alpha_2 \mathsf{K}_{\mathrm{rel}}(X) + \alpha_3 \mathsf{K}_{\mathrm{cl}}(X) + \alpha_4 \mathsf{K}_{\mathrm{res}}(X), \qquad \alpha_i > 0.$$

Lemma 2. *For every $X \in \mathcal{X}$,*

$$0 \le \mathsf{K}(X) \le \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4.$$

Proof. Each component lies in $[0,1]$, and each weight is positive.

Remark 2. Total coherence measures coherence present in the state. It does not yet measure whether that coherence remains navigable, supportable, recoverable, or operationally exploitable.

That distinction is decisive. A state may contain substantial residual structure while being entropically degraded if that structure is no longer accessible.

7 Accessibility as a structural filter

Definition 11 (Accessibility components)**.** Let

$$\mathsf{A}_{\mathrm{nav}},\ \mathsf{A}_{\mathrm{sup}},\ \mathsf{A}_{\mathrm{rec}},\ \mathsf{A}_{\mathrm{act}} : \mathcal{X} \to [0,1]$$

denote navigability, supportability, recoverability, and activatability of coherence respectively.

Definition 12 (Coherence accessibility operator)**.** Define the accessibility operator

$$\mathsf{A}_c : \mathcal{X} \to [0,1]$$

by the normalised weighted mean

$$\mathsf{A}_c(X) = \frac{\beta_1 \mathsf{A}_{\mathrm{nav}}(X) + \beta_2 \mathsf{A}_{\mathrm{sup}}(X) + \beta_3 \mathsf{A}_{\mathrm{rec}}(X) + \beta_4 \mathsf{A}_{\mathrm{act}}(X)}{\beta_1 + \beta_2 + \beta_3 + \beta_4}, \qquad \beta_i > 0.$$

Axiom 1 (Accessibility monotonicity under structural degradation)**.** *Let $X \rightsquigarrow_A Y$. If the transition from X to Y weakens one or more of navigability, supportability, recoverability, or activatability, while none of the remaining accessibility components increases sufficiently to compensate for that weakening in the weighted mean, then*

$$\mathsf{A}_c(Y) \le \mathsf{A}_c(X).$$

If the weighted degradation is strict, then

$$\mathsf{A}_c(Y) < \mathsf{A}_c(X).$$

Remark 3. This axiom does not force accessibility to decrease on every admissible transition. It states only that accessibility is monotone under uncompensated structural degradation of the accessibility components. This is the entropy-relevant class.

Lemma 3. *For every $X \in \mathcal{X}$,*

$$0 \leq \mathsf{A}_c(X) \leq 1.$$

Proof. The operator is a normalised positive weighted mean of quantities in $[0, 1]$.

Definition 13 (Accessible coherence)**.** Define accessible coherence by

$$\mathsf{K}_{\mathrm{acc}}(X) := \mathsf{A}_c(X)\,\mathsf{K}(X).$$

Lemma 4. *For every $X \in \mathcal{X}$,*

$$0 \leq \mathsf{K}_{\mathrm{acc}}(X) \leq \mathsf{K}(X).$$

Proof. Since $0 \leq \mathsf{A}_c(X) \leq 1$, multiplication by $\mathsf{K}(X) \geq 0$ gives the result.

This inequality already formalises one of the key claims of the paper: the coherence relevant to entropy is always a restricted subset of the total coherence present.

Proposition 1 (Accessible coherence monotonicity under uncompensated accessibility loss)**.** *Let $X \rightsquigarrow_A Y$. If*

$$\mathsf{A}_c(Y) \leq \mathsf{A}_c(X) \qquad \textit{and} \qquad \mathsf{K}(Y) \leq \mathsf{K}(X),$$

then

$$\mathsf{K}_{\mathrm{acc}}(Y) \leq \mathsf{K}_{\mathrm{acc}}(X).$$

If either inequality is strict and the other is non-increasing, then

$$\mathsf{K}_{\mathrm{acc}}(Y) < \mathsf{K}_{\mathrm{acc}}(X).$$

8 Closure-accessible coherence

Accessible coherence is still too broad for the strongest Darke-Theoretic entropy law. A further gate is needed to isolate coherence that remains not merely usable, but closure-bearing.

Definition 14 (Closure gate)**.** Let

$$\Gamma : [0, \infty) \to [0, 1]$$

be a monotone nondecreasing closure-gating function.

Remark 4. The function Γ may be taken as a hard threshold, a smooth sigmoid, or any monotone saturating profile depending on how sharply the regime transition is to be modelled.

Definition 15 (Closure-accessible coherence)**.** Define closure-accessible coherence by

$$\mathsf{K}_{\mathrm{cl,acc}}(X) := \Gamma(\mathsf{Cl}(X))\,\mathsf{K}_{\mathrm{acc}}(X).$$

Lemma 5. *For every $X \in \mathcal{X}$,*

$$0 \leq \mathsf{K}_{\mathrm{cl,acc}}(X) \leq \mathsf{K}_{\mathrm{acc}}(X) \leq \mathsf{K}(X).$$

Proof. Since $\Gamma(\mathsf{Cl}(X)) \in [0, 1]$, multiplication preserves the stated bounds.

This object is the real load-bearing quantity of the paper. Entropy will be defined from it, not from naive disorder.

9 Darke entropy functional

Definition 16 (General Darke entropy)**.** Let

$$\Phi : (0, \infty) \to \mathbb{R}$$

be strictly decreasing. Define Darke entropy by

$$\mathsf{S}_D(X) := \Phi\big(\mathsf{K}_{\mathrm{cl,acc}}(X)\big)$$

whenever $\mathsf{K}_{\mathrm{cl,acc}}(X) > 0$.

Definition 17 (Linear model)**.** A linear model is given by

$$\mathsf{S}_D(X) := K_\star - \lambda\, \mathsf{K}_{\mathrm{cl,acc}}(X), \qquad K_\star > 0,\ \lambda > 0.$$

Definition 18 (Logarithmic model)**.** A logarithmic model is given by

$$\mathsf{S}_D(X) = -\lambda \log\left(\frac{\mathsf{K}_{\mathrm{cl,acc}}(X)}{K_\star}\right), \qquad K_\star > 0,\ \lambda > 0.$$

Remark 5. The linear form is algebraically simple. The logarithmic form has the advantage that multiplicative coherence losses become additive entropy increments and collapse of closure-accessible coherence drives entropy upward sharply.

Definition 19 (Entropy increment)**.** For a transition $X \to Y$, define

$$\Delta\mathsf{S}_D(X, Y) := \mathsf{S}_D(Y) - \mathsf{S}_D(X).$$

In the linear model,

$$\Delta\mathsf{S}_D(X, Y) = -\lambda\big(\mathsf{K}_{\mathrm{cl,acc}}(Y) - \mathsf{K}_{\mathrm{cl,acc}}(X)\big).$$

This immediately yields the paper's central equivalence.

Theorem 1 (Entropy-coherence equivalence)**.** *Let Φ be strictly decreasing. Then for any states X, Y with positive closure-accessible coherence,*

$$\Delta\mathsf{S}_D(X, Y) > 0 \iff \mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X).$$

Proof. Since Φ is strictly decreasing,

$$\mathsf{S}_D(Y) > \mathsf{S}_D(X) \iff \Phi\big(\mathsf{K}_{\mathrm{cl,acc}}(Y)\big) > \Phi\big(\mathsf{K}_{\mathrm{cl,acc}}(X)\big) \iff \mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X).$$

Thus entropy increase is not primitive disorder. It is precisely loss of closure-accessible coherence.

10 Why disorder is inadequate

The usual disorder picture fails because entropy does not merely track how scattered or irregular a state appears. It tracks how much structurally usable coherence remains accessible.

Proposition 2 (Structure-rich yet entropically degraded states)**.** *There exist states $X, Y \in \mathcal{X}$ such that*

$$\mathsf{K}(Y) \geq \mathsf{K}(X) \qquad \textit{but} \qquad \mathsf{S}_D(Y) > \mathsf{S}_D(X).$$

Proof. Choose Y such that its total coherence is preserved or even increased, but its accessibility and closure gate are sufficiently reduced:

$$\Gamma(\mathsf{Cl}(Y))\,\mathsf{A}_c(Y)\,\mathsf{K}(Y) < \Gamma(\mathsf{Cl}(X))\,\mathsf{A}_c(X)\,\mathsf{K}(X).$$

Then

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X),$$

and hence, by the entropy-coherence equivalence theorem,

$$\mathsf{S}_D(Y) > \mathsf{S}_D(X).$$

This is a decisive result. It shows that entropy can rise even while substantial structure remains. Entropy is therefore not fundamentally the same thing as visual or descriptive disorder.

Corollary 1. *A state may be coherence-rich yet entropically degraded if its coherence is inaccessible, irrecoverable, unsupported, or closure-deficient.*

11 Irreversibility and recovery burden

Entropy increase becomes structurally meaningful when lost accessible coherence cannot be restored admissibly.

Definition 20 (Recovery burden)**.** For a transition from X to Y, define the reverse recovery burden

$$\mathsf{B}_{\mathrm{rec}}(Y \to X; \ell)$$

as the burden required to restore the closure-accessible coherence lost in passing from X to Y.

Definition 21 (Recovery reserve)**.** Define the recovery reserve by

$$\Pi_{\mathrm{rec}}(Y \to X) := \mathsf{S}(Y \to X; \ell) - \mathsf{B}_{\mathrm{rec}}(Y \to X; \ell).$$

Definition 22 (Entropic irreversibility)**.** A positive entropy increment

$$\Delta\mathsf{S}_D(X, Y) > 0$$

is structurally irreversible on the admissible sector if

$$\Pi_{\mathrm{rec}}(Y \to X) < \pi_{\min}.$$

Proposition 3 (Irreversible entropy increase)**.** *If*

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X) \qquad \textit{and} \qquad \Pi_{\mathrm{rec}}(Y \to X) < \pi_{\min},$$

then

$$\mathsf{S}_D(Y) > \mathsf{S}_D(X)$$

and the entropy increase is structurally irreversible on the admissible sector.

Proof. The first inequality implies entropy increase by the entropy-coherence equivalence theorem. The second inequality states that admissible recovery of the lost closure-accessible coherence is unavailable. Therefore the increase is structurally irreversible on the admissible sector.

This is a more precise statement than the usual slogan that entropy tends to increase. Entropy increases irreversibly when closure-accessible coherence is lost beyond admissible recovery.

12 Directional asymmetry and entropy growth

The irreversibility logic can be sharpened further.

Definition 23 (Support surplus)**.** For a transition $X \to Y$, define

$$\Sigma(X, Y; \ell) := \Pi(X \to Y).$$

Definition 24 (Directional asymmetry)**.** Define

$$\Omega(X, Y; \ell) := \Sigma(X, Y; \ell) - \Sigma(Y, X; \ell).$$

Remark 6. Positive Ω indicates that the forward transition is structurally favoured over the reverse at the reserve level.

Proposition 4 (Asymmetry-to-recovery failure bridge)**.** *Assume*

$$\Pi(X \to Y) \geq \pi_{\min}$$

and

$$\Omega(X, Y; \ell) > \Pi(X \to Y) - \pi_{\min}.$$

Then

$$\Pi(Y \to X) < \pi_{\min}.$$

Proof. By definition,

$$\Omega(X, Y; \ell) = \Pi(X \to Y) - \Pi(Y \to X).$$

If

$$\Omega(X, Y; \ell) > \Pi(X \to Y) - \pi_{\min},$$

then rearranging gives

$$\Pi(Y \to X) < \pi_{\min}.$$

Definition 25 (Recovery-aligned reverse sector)**.** A sector of the state space is recovery-aligned if, for reverse transitions attempting to restore the closure-accessible coherence lost in a forward admissible transition $X \rightsquigarrow_A Y$, the reverse reserve and the recovery reserve coincide:

$$\Pi(Y \to X) = \Pi_{\text{rec}}(Y \to X).$$

Remark 7. On a recovery-aligned reverse sector, directional asymmetry statements expressed in terms of reverse reserve become directly interpretable as statements about the admissible recoverability of lost closure-accessible coherence.

Proposition 5 (Asymmetry-to-entropic-irreversibility bridge on recovery-aligned sectors)**.** *Let $X \rightsquigarrow_A Y$ lie in a recovery-aligned reverse sector. If*

$$\Pi(X \to Y) \geq \pi_{\min} \qquad \textit{and} \qquad \Omega(X, Y; \ell) > \Pi(X \to Y) - \pi_{\min},$$

then

$$\Pi_{\text{rec}}(Y \to X) < \pi_{\min}.$$

Hence any positive entropy increment from X to Y is structurally irreversible on that sector.

Proof. By the asymmetry-to-recovery failure bridge,

$$\Pi(Y \to X) < \pi_{\min}.$$

Since the sector is recovery-aligned,

$$\Pi_{\mathrm{rec}}(Y \to X) = \Pi(Y \to X),$$

so

$$\Pi_{\mathrm{rec}}(Y \to X) < \pi_{\min}.$$

If in addition the forward transition produces

$$\Delta \mathsf{S}_D(X, Y) > 0,$$

then the resulting entropy increase is structurally irreversible by the definition of entropic irreversibility.

13 Entropy theorem chain

We now collect the principal results.

Theorem 2 (Entropy increase under coherence loss)**.** *If*

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X),$$

then

$$\mathsf{S}_D(Y) > \mathsf{S}_D(X).$$

Proof. Immediate from strict monotonic decrease of Φ.

Theorem 3 (Entropy preservation under coherence invariance)**.** *If*

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) = \mathsf{K}_{\mathrm{cl,acc}}(X),$$

then

$$\mathsf{S}_D(Y) = \mathsf{S}_D(X).$$

Proof. Immediate from the functional definition of S_D.

Theorem 4 (Entropy increase without total coherence loss)**.** *If*

$$\mathsf{K}(Y) \geq \mathsf{K}(X)$$

but

$$\Gamma(\mathsf{Cl}(Y))\, \mathsf{A}_c(Y)\, \mathsf{K}(Y) < \Gamma(\mathsf{Cl}(X))\, \mathsf{A}_c(X)\, \mathsf{K}(X),$$

then

$$\mathsf{S}_D(Y) > \mathsf{S}_D(X).$$

Proof. The stated inequality is exactly the condition

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X).$$

Apply the entropy increase theorem.

Theorem 5 (Irreversible entropy theorem)**.** *If*

$$\Delta \mathsf{S}_D(X, Y) > 0 \qquad \textit{and} \qquad \Pi_{\mathrm{rec}}(Y \to X) < \pi_{\min},$$

then the entropy increase from X to Y is direction-bearing on the admissible sector.

Proof. Positive entropy increment means closure-accessible coherence has fallen. Failure of recovery reserve means admissible restoration of the lost coherence is unavailable. Hence the increase is direction-bearing on the admissible sector.

14 Relation to thermodynamic entropy

The present framework does not deny thermodynamics. It relocates it.

Let

$$\Omega_{\text{micro}}(X)$$

denote the number of admissible micro-configurations compatible with an effective macro-regime associated with X. Standard thermodynamic entropy takes the form

$$S_{\text{th}}(X) = k_B \log \Omega_{\text{micro}}(X).$$

Darke Theory reinterprets such expressions as coarse-grained macrostatistical proxies for a deeper coherence-accessibility law.

Proposition 6 (Macrostatistical projection principle)**.** *Suppose a sector of the state space satisfies:*

i. coarse-graining is stable,

ii. closure-accessible coherence decreases monotonically with effective admissible microstate multiplicity,

iii. recovery costs are well-approximated statistically at the macro level.

Then Darke entropy admits an effective macrostatistical projection of the form

$$\mathsf{S}_D(X) \sim \lambda \log \Omega_{\text{eff}}(X) + \text{const},$$

for a suitable effective multiplicity $\Omega_{\text{eff}}(X)$*.*

Proof. If, on the relevant sector,

$$\mathsf{K}_{\text{cl,acc}}(X) \propto \Omega_{\text{eff}}(X)^{-1},$$

then in the logarithmic entropy model,

$$\mathsf{S}_D(X) = -\lambda \log\left(\frac{\mathsf{K}_{\text{cl,acc}}(X)}{K_\star}\right) = \lambda \log \Omega_{\text{eff}}(X) + \text{const}.$$

Remark 8. This does not make thermodynamic entropy false. It makes it derivative. Thermodynamic entropy becomes an effective projection valid in macrostatistical sectors where accessibility loss is well approximated by multiplicity growth.

15 Entropy and memory

Entropy and memory are related, but they are not identical.

Definition 26 (Structural memory)**.** Let

$$\mathsf{M}(X) = \alpha_m \mathsf{Cl}(X) + \beta_m \mathsf{Res}(X) + \gamma_m \mathsf{Br}(X), \qquad \alpha_m, \beta_m, \gamma_m > 0,$$

with residue and branch-loss as previously defined in the wider Darke-Theoretic corpus.

Remark 9. Memory tracks persistent structural commitment and excluded alternatives. Entropy tracks loss of closure-accessible coherence. These are linked under irreversibility, but they are not the same quantity.

Proposition 7 (Sector-conditional entropy-memory coupling)**.** *Let $\mathcal{U} \subseteq \mathcal{X} \times \mathcal{X}$ be an irreversible admissible sector such that for every $(X, Y) \in \mathcal{U}$:*

i. $\Delta\mathsf{S}_D(X, Y) > 0$ *implies loss of closure-accessible coherence,*

ii. that loss deposits non-negative residue or produces non-negative branch-loss,

iii. there exists a sector constant $\mu_{\mathcal{U}} > 0$ bounding below the induced memory gain per unit entropy gain.

Then for all $(X, Y) \in \mathcal{U}$ with

$$\Delta\mathsf{S}_D(X, Y) > 0,$$

one has

$$\Delta\mathsf{M}(X, Y) \geq \mu_{\mathcal{U}}\, \Delta\mathsf{S}_D(X, Y).$$

Proof. By assumptions (i) and (ii), every positive entropy increment on $\mathcal{U}$ induces non-negative increments in one or both of the memory-bearing components, residue and branch-loss. Since the memory functional is

$$\mathsf{M}(X) = \alpha_m \mathsf{Cl}(X) + \beta_m \mathsf{Res}(X) + \gamma_m \mathsf{Br}(X),$$

with positive coefficients, these component increases induce a positive memory increment. Assumption (iii) supplies a uniform lower sector bound $\mu_{\mathcal{U}}$ relating memory gain to entropy gain. Hence

$$\Delta\mathsf{M}(X, Y) \geq \mu_{\mathcal{U}}\, \Delta\mathsf{S}_D(X, Y).$$

Remark 10. This proposition is sector-conditional, not universal. It applies where entropy increase is carried by irreversible coherence loss that necessarily deposits residue or excludes admissible branch structure.

Corollary 2. *Entropy increase tends to accompany structural memory growth on irreversible admissible sectors, without reducing one concept to the other.*

16 Master entropy law

The entire construction may be compressed into a single functional law.

Theorem 6 (Master Darke entropy law)**.** *Let Φ be strictly decreasing and let Γ be monotone nondecreasing into $[0, 1]$. Then Darke entropy may be written as*

$$\mathsf{S}_D(X) = \Phi\Big(\Gamma(\mathsf{Cl}(X)) \cdot \mathsf{A}_c(X) \cdot \mathsf{K}(X)\Big).$$

Proof. By definition,

$$\mathsf{K}_{\mathrm{acc}}(X) = \mathsf{A}_c(X)\, \mathsf{K}(X)$$

and

$$\mathsf{K}_{\mathrm{cl,acc}}(X) = \Gamma(\mathsf{Cl}(X))\, \mathsf{K}_{\mathrm{acc}}(X).$$

Substitution into the general entropy definition gives the result.

Remark 11. This law captures the entire paper in one line:

i. total coherence says how much coherence is present,

ii. accessibility says how much remains usable,

iii. closure gating says how much of that usable coherence is closure-bearing,

iv. the entropy transform reverses the ordering so that entropy increases as this product falls.

17 Interpretive consequences

The formal construction above yields several important consequences.

17.1 Entropy is not disorder

A state can be highly structured and yet entropically degraded if its coherence has become inaccessible, unsupported, irrecoverable, or closure-deficient.

17.2 Entropy is not mere multiplicity

Multiplicity-based entropy is a macrostatistical shadow that applies only where accessibility loss is well represented by effective microstate proliferation.

17.3 Irreversibility is recovery failure

Entropy increase becomes direction-bearing when closure-accessible coherence cannot be restored admissibly.

17.4 Residual structure need not mean low entropy

Residual coherence may remain present in a state even when entropy is high, provided that coherence is no longer accessible in the relevant structural sense.

17.5 Entropy and memory are linked but distinct

Entropy measures lost closure-accessible coherence. Memory measures persistent structural commitment. They co-vary on irreversible sectors but are not identical.

18 Scope

This paper is a foundational reconstruction of entropy, not a completed thermodynamic formalism for every physical regime. It does not attempt to derive all statistical mechanics from first principles in one step. Its claim is narrower and deeper: the foundational meaning of entropy is misplaced if treated as primitive disorder, and Darke Theory contains the machinery required to rebuild it from coherence, closure, accessibility, support, burden, and admissible recovery.

19 Conclusion

Entropy in Darke Theory is not primitive disorder.

The disorder picture is too coarse, too observer-dependent, and too ontologically weak to bear the load usually placed upon entropy. What matters structurally is not whether a state appears messy, but whether its coherence remains accessible under closure, support, burden, and recovery constraints.

This paper has rebuilt entropy accordingly.

A total coherence functional was introduced and decomposed into internal, relational, closure-bearing, and residual components. A coherence accessibility operator was defined to measure what fraction of that coherence remains navigable, supportable, recoverable, and operationally usable. Closure-accessible coherence was then obtained by gating accessible coherence through the closure functional. Darke entropy was defined as a strictly decreasing transform of this quantity. From that definition, the key result follows:

Entropy increases if and only if closure-accessible coherence decreases.

The paper then showed that entropy may increase even when total coherence is preserved or increased, thereby breaking the naive equation of entropy with disorder. Irreversibility was reconstructed in terms of inadmissible recovery of lost coherence. Thermodynamic entropy was reinterpreted as a macrostatistical projection of the deeper coherence-accessibility law. Finally, entropy increase was related to structural memory growth without collapsing the two into identity.

The compressed statement of the paper is therefore:

> **Entropy in Darke Theory is the inverse image of persistent accessible coherence under structural constraint.**

Entropy is not what remains when order disappears. It is what grows when closure-bearing coherence can no longer be accessed, supported, or recovered.

A Appendix: boundedness and monotonicity summary

For convenience, the principal structural bounds and monotonicities used in the paper are collected here.

1. Closure subfunctionals satisfy
$$I_{\text{self}},\ I_{\text{rec}},\ I_{\text{supp}},\ I_{\text{pert}} \in [0,1].$$
2. Total coherence components satisfy
$$\mathsf{K}_{\text{int}},\ \mathsf{K}_{\text{rel}},\ \mathsf{K}_{\text{cl}},\ \mathsf{K}_{\text{res}} \in [0,1].$$
3. Accessibility components satisfy
$$\mathsf{A}_{\text{nav}},\ \mathsf{A}_{\text{sup}},\ \mathsf{A}_{\text{rec}},\ \mathsf{A}_{\text{act}} \in [0,1].$$
4. Closure gate satisfies
$$0 \le \Gamma(\mathsf{Cl}(X)) \le 1.$$
5. Therefore
$$0 \le \mathsf{K}_{\text{cl,acc}}(X) \le \mathsf{K}_{\text{acc}}(X) \le \mathsf{K}(X).$$
6. Entropy transform satisfies
$$\Phi'(u) < 0$$
on its domain.
7. Recovery irreversibility occurs when
$$\Pi_{\text{rec}}(Y \to X) < \pi_{\min}.$$

B Appendix: expected monotonicity conditions

The main text does not require every monotonicity property to be derived from a single deeper law. It is useful, however, to record sufficient conditions under which the key monotonicities are naturally expected.

Proposition 8 (Expected accessibility monotonicity failure under degradation)**.** *Let $X \rightsquigarrow_A Y$. Assume that the transition from X to Y:*

i. preserves or increases latent residual structure,

ii. reduces navigability, supportability, recoverability, or activatability of coherence,

iii. does not supply compensating closure gain sufficient to offset the accessibility loss.

Then

$$\mathsf{K}_{\mathrm{cl,acc}}(Y) < \mathsf{K}_{\mathrm{cl,acc}}(X).$$

Proof. Conditions (ii) and (iii) reduce either the accessibility factor A_c or the closure gate $\Gamma(\mathsf{Cl}(\cdot))$ sufficiently that the product

$$\Gamma(\mathsf{Cl}(\cdot))\,\mathsf{A}_c(\cdot)\,\mathsf{K}(\cdot)$$

decreases, even if latent residual structure persists by (i).

Proposition 9 (Expected irreversibility under recovery overload)**.** *Suppose $X \rightsquigarrow_A Y$ and that restoration of the lost closure-accessible coherence requires:*

i. rebuilding previously excluded coherence pathways,

ii. reversing branch exclusions or residue deposition,

iii. supplying support whose recovery burden exceeds admissible reserve.

Then

$$\Pi_{\mathrm{rec}}(Y \to X) < \pi_{\min}.$$

Proof. Assumption (iii) directly states that support minus recovery burden falls below the admissible reserve threshold.

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